



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	8.17 / 10
2023	AISSCE	Delhi Model Public School, Patna	94.2%
2021	AISSE	Bishop Scott Boys School, Patna	98%

INTERNSHIPS

- AI/ML Engineer | Aristhrottle | LangGraph, PostgreSQL, LangFuse, FastAPI** [June '25 - Present]
(*Overview* : Built a production grade AI pipeline for automated story generation maintaining visual coherence across multi-panel narratives)
- Architected a multi-stage AI pipeline combining **LLM narrative** planning with Google Imagen3 API, powered by PostgreSQL for **scalable content storage**
 - Engineered a prompt deconstruction **microservice** using Gemini API to extract structured elements, enabling **dynamic and reusable narrative** generation
 - Designed a **stochastic** prompt selector applying temperature scaling and **nucleus sampling** on ranked user data, boosting creative diversity by **~25%**
 - Integrated **LangFuse** SDK to trace **LangGraph** executions, monitor token usage and latency, aiding debugging and **A/B testing** of prompt strategies
- Undergraduate Researcher | KLIV - Kharagpur Learning, Imaging and Visualization Research Group | IIT Kharagpur** [Mar '25 - Pr.]
- Conducting research and publication efforts in agentic AI for medical applications under Prof. Debdoot Sheet and Mr. Sista Raviteja, leveraging LLMs
 - Actively developing real-world deployment of the RAD-Agent for automated report & EHR generation for radiologist and doctors in hospital workflows

PROJECTS

- Vision-guided RAG pipeline | Self Project | LlamaParse, LangChain** [June '25]
(*Overview*: Developed a Vision-Guided RAG pipeline, inspired by recent research, boosting chunking accuracy by ~10% over vanilla RAG)
- Developed a **multimodal** data processing pipeline using **LlamaParse** and **LangChain**, producing **5x** more granular chunks for precise information retrieval
 - Designed specialized **prompt templates** for the LLMs, preserving **mathematical notation** and **logical structure** within complex scientific documents
 - Resolved LlamaParse limitations with scanned PDFs by integrating **Tesseract OCR**, ensuring **robust** text extraction from **low-quality** scanned images
 - Optimized resources by implementing **dynamic batching** and **context-aware chunking** strategies, overcoming the model's **output token limit** constraints
- AI Traffic Analyst | Self Project | YOLOv8, DeepSORT, Prophet** [May '25]
(*Overview*: Built an AI traffic analyst pipeline with YOLOv8, DeepSort & Prophet for vehicle tracking, aggregation, and congestion forecasting)
- Engineered **computer vision** pipeline processing **24** hours of **Bellevue** dataset video, integrating **YOLOv8** and **SORT** for vehicle detection and tracking
 - Enhanced tracking accuracy by implementing **DeepSORT** algorithm, resolving **ID-switching** issues in dense traffic scenarios with deep-sort-realtime
 - Constructed **time-series aggregation** framework compiling vehicle counts into **5-minute** intervals, generating **structured CSV** dataset for modeling
 - Formulated **Prophet** forecasting model predicting **traffic congestion** over **3-hour** horizon, identifying daily seasonality trends with **Plotly** visualization

PUBLICATIONS

- RAD-Agent: An Agentic AI for Voice-Driven Dataset Automation** [Under Review]
(*Overview*: Built an agentic AI system to generate medical reports directly from radiologist dictations, while ensuring good clinical accuracy)
- Built a **five-stage agentic AI pipeline** converting radiologist voice to structured datasets, cutting costs by **70–80%** over traditional manual annotation
 - Designed hybrid medical entity recognition using **RadGraph**, configurable **NER**, and regex patterns, maintaining **F-score** above **0.80** across modalities
 - Implemented **RAG-enhanced** form population with **FAISS** vector store and dual-LLM validation framework, ensuring structural accuracy exceeding **0.85**
 - Architected privacy-by-design framework with **WhisperX** transcription, **Microsoft Presidio** PHI removal, and **AES-GCM** encryption for **HIPAA** compliance

COMPETITION/CONFERENCE

- Budget optimization | Data Analytics General Championship 2025 | Bronze Winning Team | IIT Kharagpur** [Mar '25]
(*Objective*: To leverage advanced data analytics to optimize a multi-million dollar Electromart's marketing budget and maximize it's revenue)
- Engineered a **Marketing Mix Model** with Meta's **Robyn** to optimize an **\$18.7M** budget, driving a **31.9%** revenue uplift across 3 marketing channels
 - Formulated predictive models and a response decomposition waterfall analysis, validating forecast accuracy against actual revenue (**p-value <0.05**)
 - Identified key **revenue drivers** by analyzing customer behavior for peak times (**4-5 PM**) and optimized **ROAS** via diminishing returns modeling analysis
- Employee Selection Engine | Open Soft General Championship 2025 | Bronze Winning Team | Azad Hall, IIT Kharagpur** [Apr '25]
(*Objective*: Develop a analytics framework that proactively identifies at-risk employees using predictive models and integrated HR data)
- Architected predictive analytics pipeline achieving **87%** accuracy in **at-risk** employee identification by integrating 6 HR datasets with **MICE imputation**
 - Engineered custom **Temporal Decay** Algorithm reducing false positives by **15%** through exponential decay prioritization of recent behavioral patterns
 - Implemented **AHP** framework with **CR<0.1** consistency validation, automating dataset weight justification through **eigenvalue** decomposition methods

SKILLS AND EXPERTISE

Languages and Frameworks: Python, C++, Javascript, ReactJs, Node.js, PyTorch, Keras, R*, ExpressJs (* elementary proficiency)
Databases & Vector DB: PostgreSQL, MongoDB (with moongose ODM), FAISS, ChromaDB, Pinecone
Tools &Additional Skills: NumPy, Pandas, Scikit-Learn, Git, GitHub, HF, Axios, OpenCV*, LangChain, LlamaIndex , Prophet, LangGraph* , LangFuse*

COURSEWORK INFORMATION

Programming and Data Structures* | Probability and Statistics | Linear Algebra, Numerical and Complex Analysis | Advanced Calculus | Signals and Systems* | File Organization and Database Systems Lab | Digital Electronics* | Intro to AI/ML (* contains Lab component)

EXTRA CURRICULAR ACTIVITIES

- Represented Azad Hall in Interhall General Championship 2025 as a core football team member, demonstrating teamwork and athletic discipline
- Participated in National Cultural Appreciation (NCA) Instrumental program, under Academy of Classical and Folk Arts, IIT Kharagpur